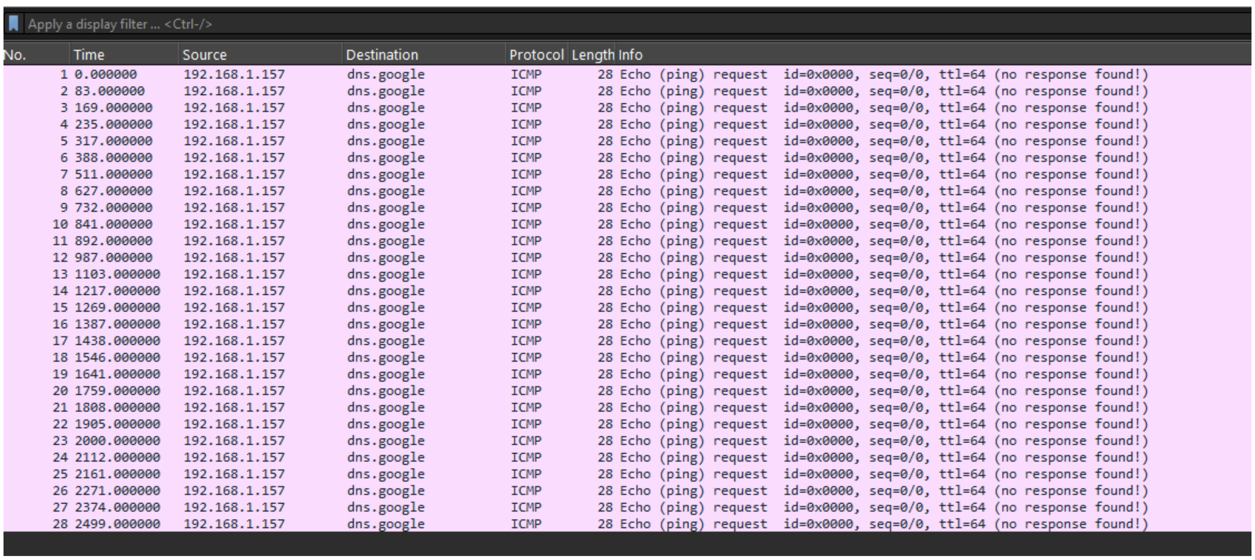
Pipeline Project: Silent Signal

- Challenge Objective: Find the flag hidden within the pcap.

Recon

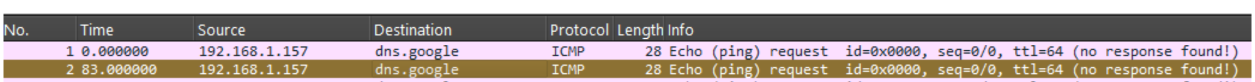
The first thing you do when given a pcap file is try to open it in Wireshark and see what you come across. When you do that, you see a bunch of ICMP traffic:


Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
2	83.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
3	169.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
4	235.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
5	317.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
6	388.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
7	511.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
8	627.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
9	732.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
10	841.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
11	892.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
12	987.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
13	1103.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
14	1217.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
15	1269.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
16	1387.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
17	1438.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
18	1546.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
19	1641.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
20	1759.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
21	1808.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
22	1905.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
23	2000.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
24	2112.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
25	2161.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
26	2271.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
27	2374.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)
28	2499.000000	192.168.1.157	dns.google	ICMP	28	Echo (ping) request id=0x0000, seq=0/0, ttl=64 (no response found!)

Solve

First thing you are going to want to do is look at the second ICMP packet in the capture. The trick here is in the time deltas. You can't have time delta without a previous packet.



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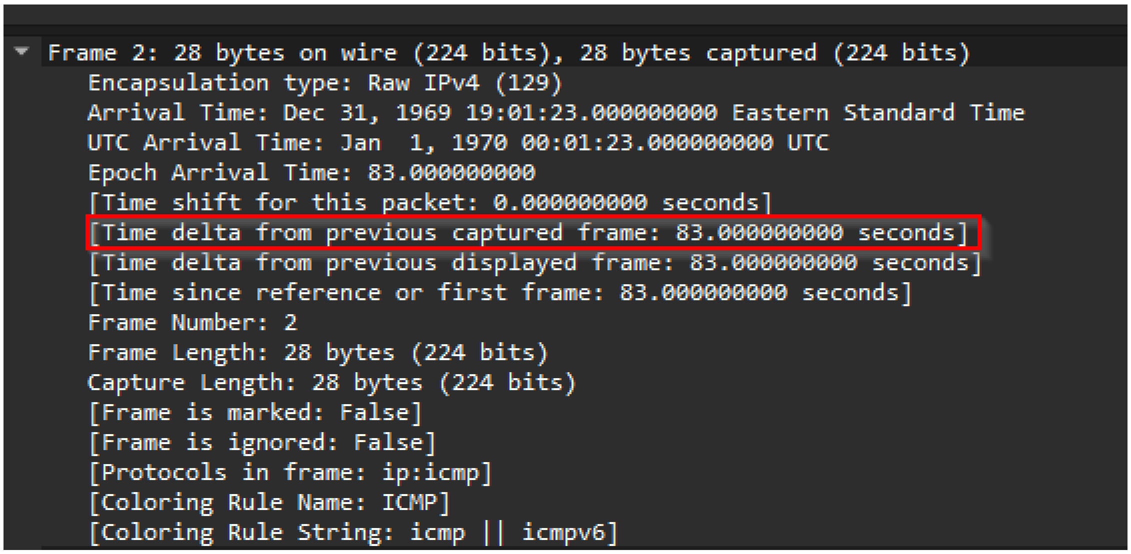
Next, you'll want to go down into the packet information, specifically the Frame dropdown. This has a lot of information on the size of the packet, timestamping, what protocols are being used, etc. For this challenge, we will want to specifically look at the time delta information.

If we look at the first packet, we'll see there is no time delta as you need a previous packet to calculate the time between packets being sent.

But, if we go to the second packet, we'll see a peculiar number:



From here, since each ICMP packet comes directly after another, we can just go through and check the time deltas of each packet.

These numbers together look a lot like ASCII number representations of characters. If we go through each packet and get the time delta, we get the following string of numbers:

```
83 86 66 82 71 123 116 105 109 51 95 116 114 52 118 51 108 95 118 49 97 95 112 49 110 103 125
```

Translated from ASCII to text we get the following flag:

```
SVBRG{tim3_tr4v3l_v1a_p1ng}
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